2D Ideal MHD Solver on the Orszag-Tang Vortex

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Outline

1. Introduction and Problem Motivation
2. Mathematical Model
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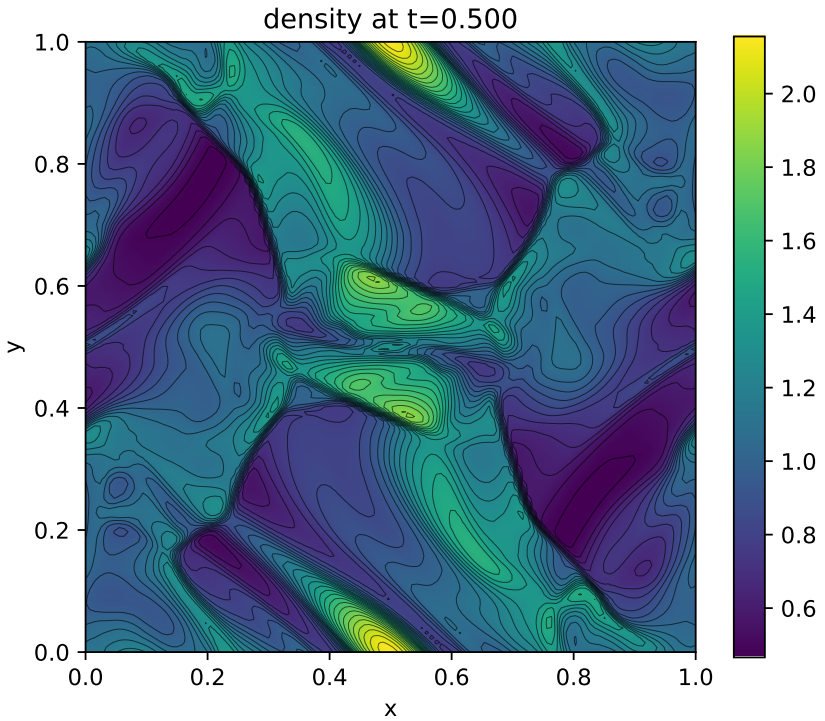
Why MHD

- A conducting fluid in motion is a distributed electric current; the *Lorentz force* $\mathbf{J} \times \mathbf{B}$ is a body force on every cell [1, 2].
- "Frozen-in flux": in the ideal limit, field lines move with the fluid. Fluid and field are tied together everywhere.
- Valid when plasma is collisional and magnetic Reynolds number is large: solar wind, tokamaks, astrophysical jets, magnetospheres.
- New physics vs Euler:
  - *Alfven* waves—transverse, like waves on a string with magnetic tension as the restoring force.
  - *Fast/slow magnetosonic* waves—sound waves modified by magnetic pressure.
  - Maxwell stresses $-B_i B_j$ in the momentum flux.
  - The no-monopoles constraint $\nabla \cdot B = 0$.

Project goal. Build a 2D ideal MHD finite-volume solver in Python and validate it on the Orszag-Tang vortex.

Why Orszag-Tang

- Canonical 2D MHD benchmark.
- All-sinusoidal ICs on $[0, 1]^2$, periodic on every side.
- No geometry, no density jump, no outflow BCs.
- If the result is wrong, it is the solver, not the setup.
- Reference: [3, 4].



Ideal MHD in conservative form

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0, \quad U = (\rho,\ \rho\mathbf{v},\ \mathbf{B},\ E)^T$$

with total energy $E = p/(\gamma - 1) + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2$ and total pressure $p_T = p + \frac{1}{2}|\mathbf{B}|^2$ [5].

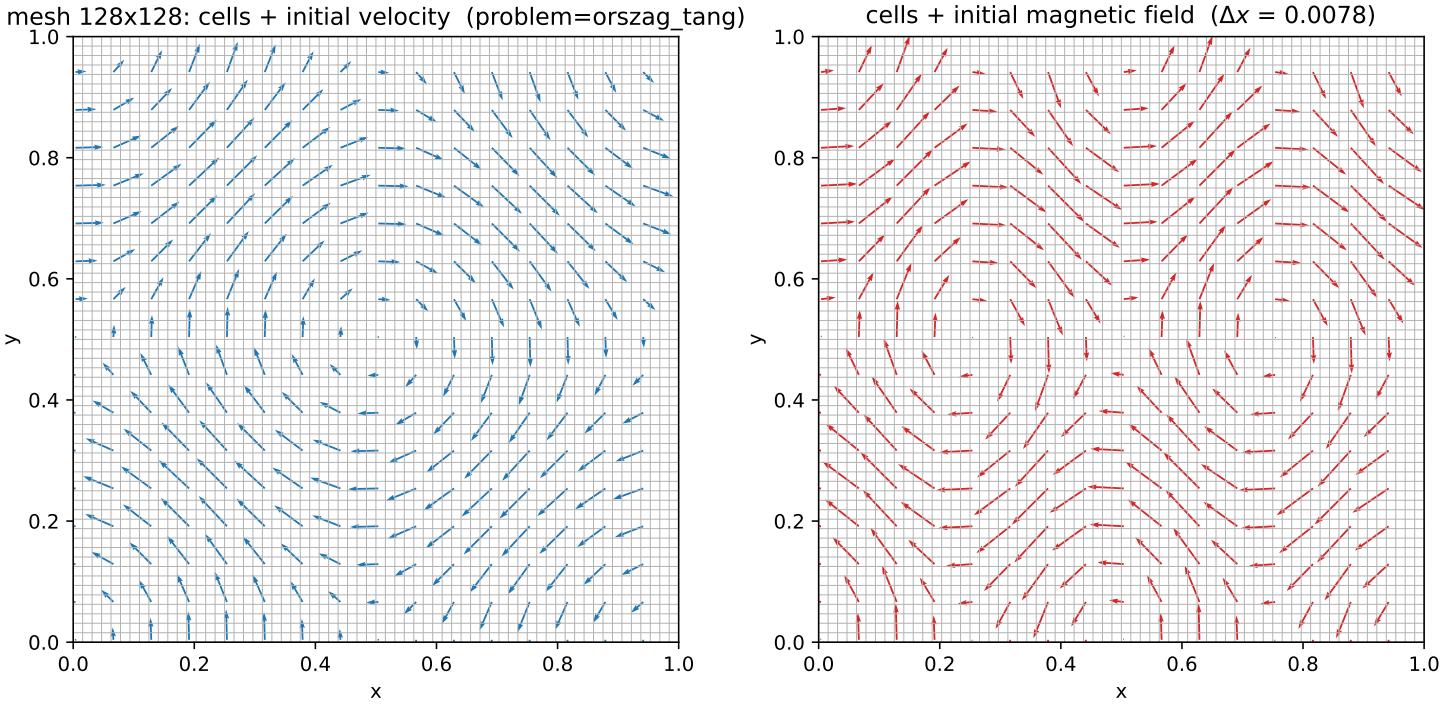
- Eight conserved variables; all three velocity and field components carried even in 2D.
- New vs Euler: *magnetic pressure* $\frac{1}{2}|B|^2$ inside p_T, *magnetic tension* along field lines via $-B_iB_j$, and *magnetic energy* $\frac{1}{2}|B|^2$ in E.
- Induction-equation flux $uB_y - vB_x$ is the z-component of the motional electric field $-\mathbf{v} \times \mathbf{B}$: this is what drags field lines along with the flow.
- Seven-wave fan: two fast magnetosonic, two Alfven, two slow magnetosonic, one entropy [6, 7].
- Ideal-gas closure, $\gamma = 5/3$.

The divergence constraint

- $\nabla \cdot B = 0$ is Gauss's law for magnetism: no magnetic monopoles, so field lines have no sources or sinks.
- Exact ideal MHD preserves this analytically; piecewise-constant Godunov updates do not—reconstructed interface fields produce a small but nonzero divergence at every cell center [8, 5].
- Consequence is direct: rewriting $(\nabla \times B) \times B = (B \cdot \nabla)B - \frac{1}{2}\nabla|B|^2 - (\nabla \cdot B)B$, the last term becomes a spurious force *parallel to* B, breaks momentum conservation, and eventually blows the run up.
- Remedies: constrained transport [8, 3], Powell source terms [5], or hyperbolic GLM cleaning [9].
- **This project uses GLM**: simpler to drop into an unstaggered FVM code, only adds one scalar ψ.

Computational mesh

- Uniform 128×128 Cartesian grid, $\Delta x = \Delta y = 1/128$.
- Two ghost layers on every side (periodic fills for Orszag-Tang).
- Left: initial velocity; right: initial magnetic field.



Scheme at a glance

- 2D structured Cartesian grid, cell-centered state, two ghost layers [10, 11].
- MUSCL linear reconstruction on primitives with minmod limiter (MC, van Leer optional) [12].
- HLLD Riemann solver [7] at every interface; degenerates to HLLC for $B = 0$.
- GLM psi/Bx coupling handled inside the Riemann solver; parabolic damping applied per RK stage [9].
- SSP-RK2 time integration [13], CFL $= 0.3$.

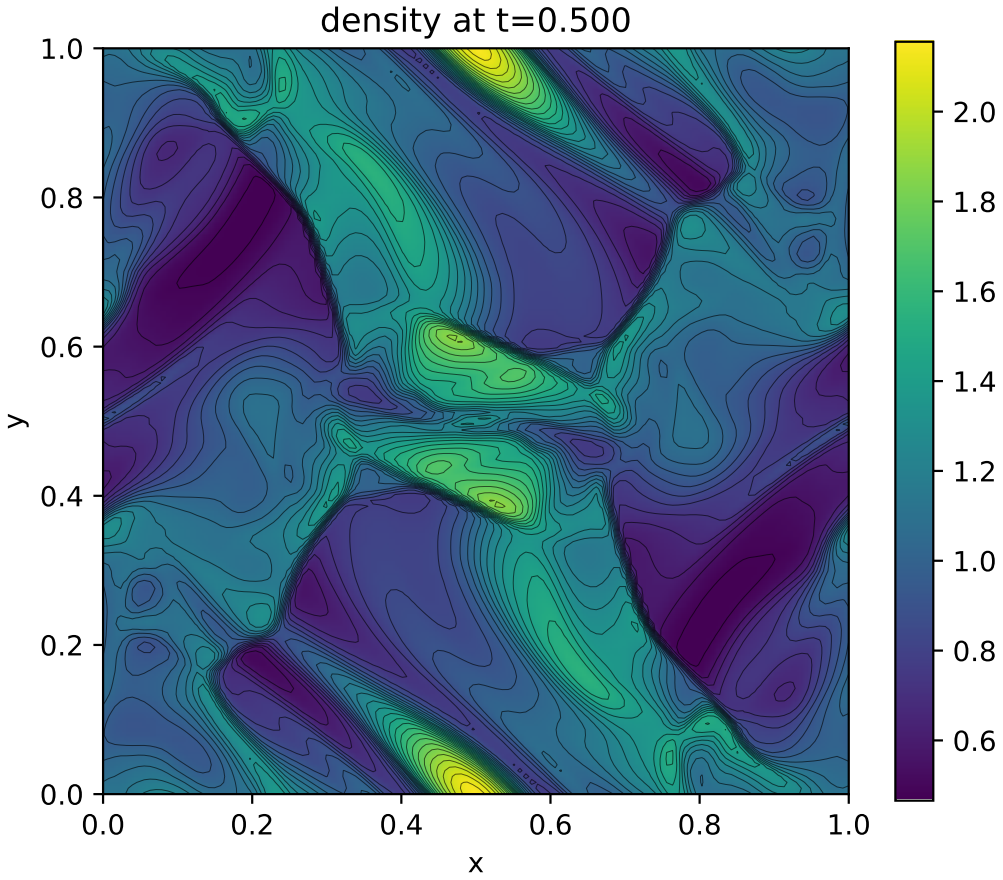
HLLD wave pattern

- Five-wave approximation to the full seven-wave MHD fan [7].
- Bounding speeds S_L, S_R: Davis-Einfeldt.
- Entropy speed S_M from jump conditions at the contact wave.
- Two starred states and two double-starred states (across the rotational Alfven discontinuities).
- Positivity-preserving for physical inputs.

L	L^*	L^{**}	R^{**}	R^*	R
S_L		S_L^*		S_M	S_R

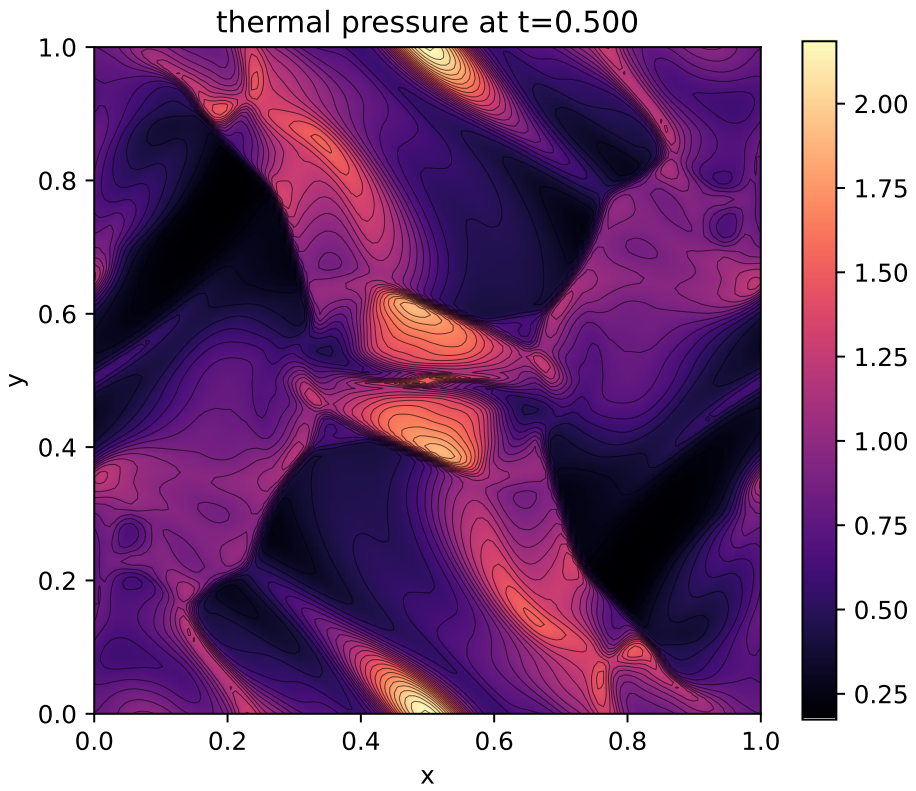
Wave regions across the x-face.

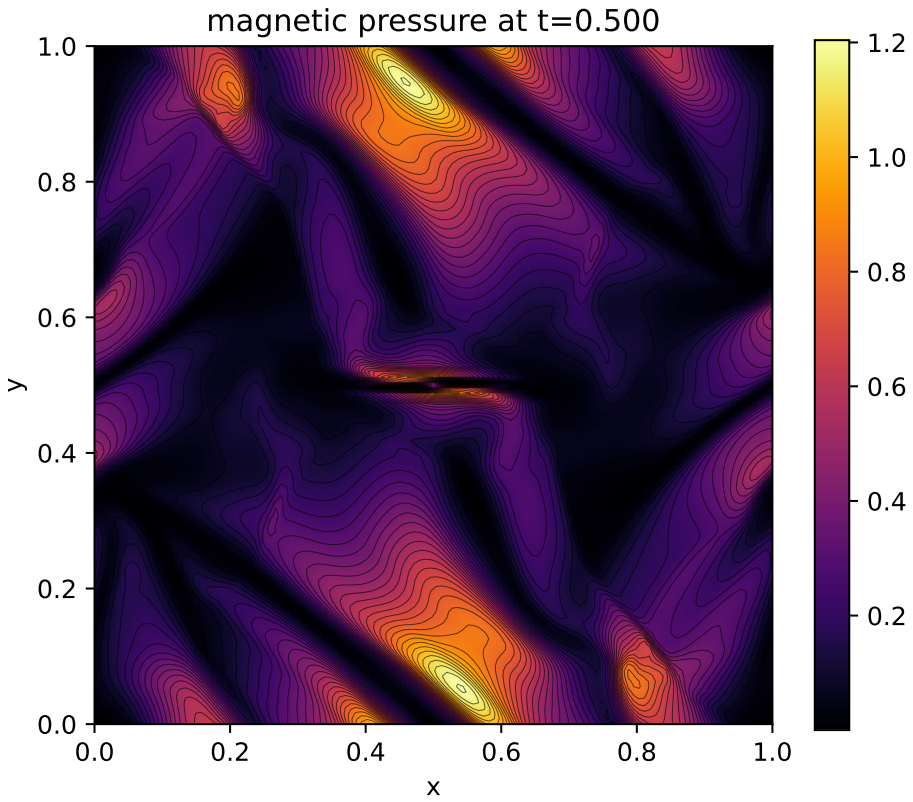
Density at t = 0.5, 128^2



- Central bow-tie current sheet.
- Four symmetric oblique shocks bounding low-density bays.
- Qualitative match with [3, 4] at the same time slice.

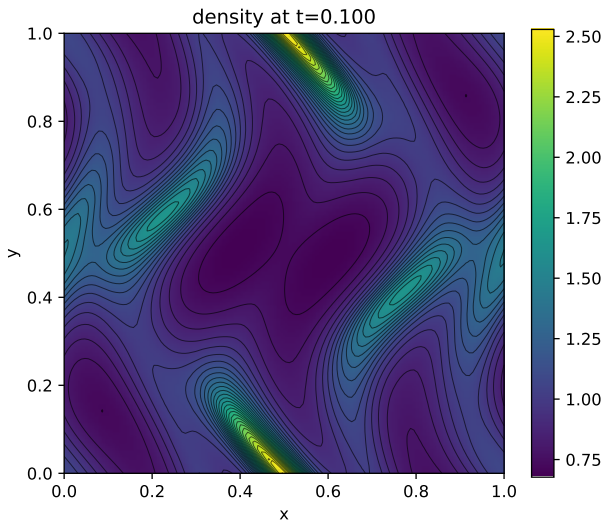
Thermal and magnetic pressure, t = 0.5



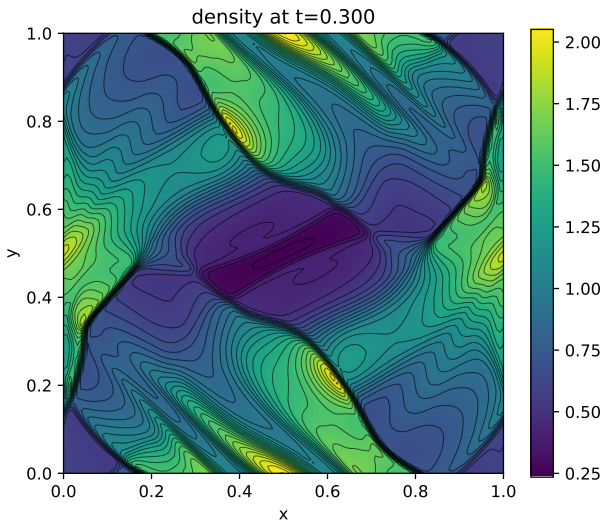


- Thermal pressure tracks density with extra shock heating near the current sheet.
- Magnetic pressure peaks at flux pile-up in the four current-sheet arms.

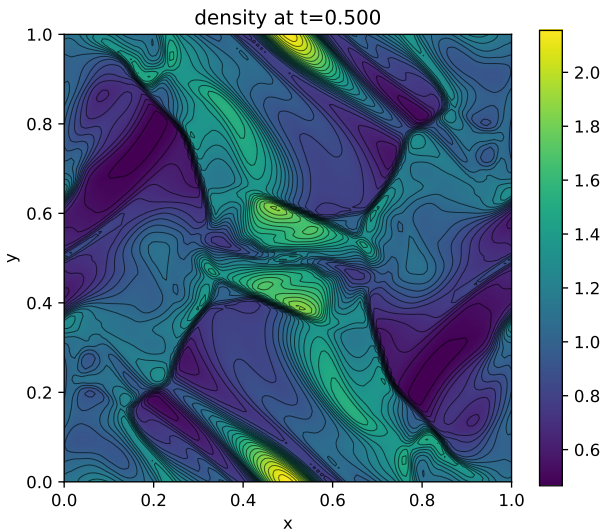
Formation of the current sheet



t = 0.1



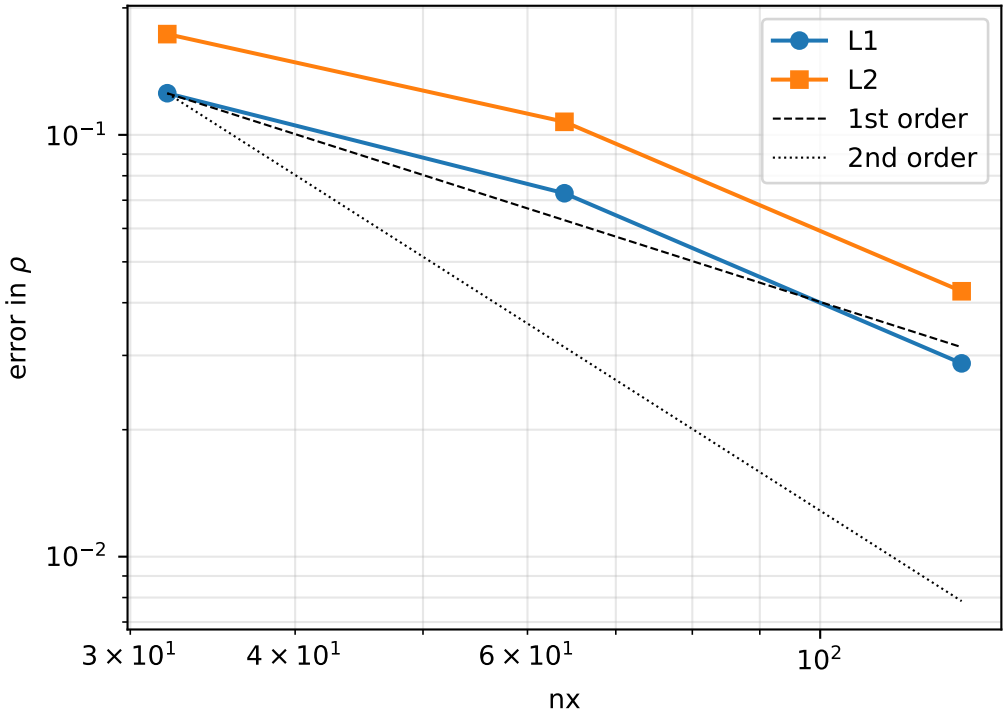
t = 0.3



t = 0.5

- t = 0.1: still smooth, the vortices are just forming.
- t = 0.3: diagonal shocks have formed; bow-tie compression starts.
- t = 0.5: fully-developed current sheet and bounding shocks.

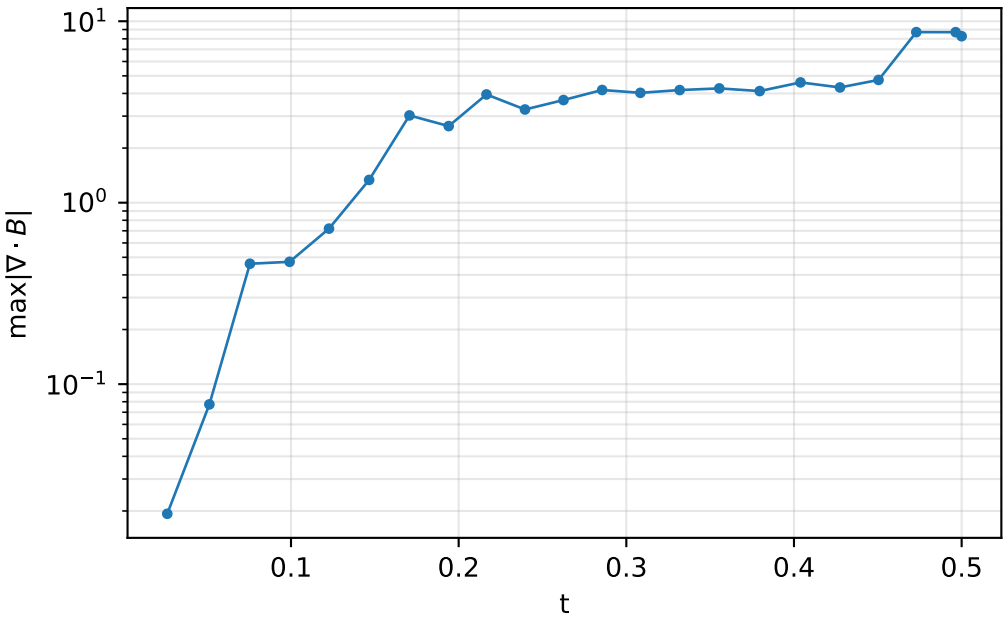
Convergence in density



nx	L^1	L^2
32	1.26e-1	1.73e-1
64	7.27e-2	1.07e-1
128	2.87e-2	4.26e-2

- $32 \to 64$: rate ≈ 0.79.
- $64 \to 128$: rate ≈ 1.33.
- Between 1st and 2nd order: expected for minmod on a shock-dominated flow [12].

Divergence control and conservation



- GLM [9] bounds $\max|\nabla \cdot B|$ once shocks form.
- Cell-centered centered differences are a pessimistic diagnostic; the GLM-natural measure is smaller.
- Mass, momentum, energy conserved to machine precision ($\sim 10^{-16}$ drift per step) – see `run.log`.

Accuracy vs limiter choice

- Observed density extrema at 128^2: $\rho_{\text{min}} \approx 0.47$, $\rho_{\text{max}} \approx 2.16$.
- Athena++ [4] with MC limiter at the same resolution: tighter extrema ($\rho_{\text{min}} \sim 0.1$, $\rho_{\text{max}} \sim 3.5$).
- Gap is almost entirely the limiter: minmod is the most diffusive TVD limiter and smears current sheets fastest [12].
- MC or van Leer would close the gap but at a cost in robustness for low-beta cells.

Why between first and second order

- MUSCL is second-order in smooth regions [12].
- Any TVD limiter drops to first order at local extrema and at shocks.
- Orszag-Tang at $t = 0.5$ is shock-dominated, so the global rate sits between 1 and 2.
- Observed: ≈ 0.8 at coarse resolutions (shock-dominated), ≈ 1.3 at fine (smooth regions start to dominate).

Limitations

- Vectorized NumPy: 128^2 runs in $\sim 7\,\mathrm{s}$; 256^2 is the practical ceiling on a laptop.
- No constrained-transport option – GLM is simpler but less strict about $\nabla \cdot B$ [8].
- 2D Cartesian only; no adaptive refinement.
- Reconstruction is linear (MUSCL) – no WENO, no higher-order time integration.

Conclusions

- Built a 2D ideal MHD solver from scratch: Godunov FVM, MUSCL-minmod, HLLD [7], GLM [9], SSP-RK2 [13].
- Orszag-Tang at 128^2 reproduces the canonical density/pressure pattern at $t = 0.5$ [3, 4].
- Mass, momentum, energy conserved to machine precision; $\max|\nabla \cdot B|$ bounded by GLM.
- Convergence rate between 1 and 2, consistent with MUSCL-minmod on a shock-dominated flow.

Natural extensions. Constrained transport; resistive MHD; 3D; AMR; Numba/JAX compilation.

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